Automated E-commerce Product Image Quality and Compliance Inspection Using Classical Image-Processing Features

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***Abstract*—Online marketplaces reject a large fraction of seller-uploaded product photographs because they are blurred, noisy, washed out, shot on a cluttered background, or badly framed. We build a lightweight inspector that flags such images and names the defect. From each image we compute ten interpretable digital-image-processing (DIP) measures—variance of the Laplacian, a median-residual noise estimate, histogram spread, border whiteness, background uniformity after Otsu segmentation with morphological clean-up, and bounding-box/centroid statistics of the largest connected component. A hand-tuned rule baseline on these measures reaches 91.3 % accuracy (macro-F1 0.915) on a held-out, product-disjoint test set of 540 images. Replacing the rules by a Random Forest trained on the same ten features raises this to 98.3 % (macro-F1 0.983), outperforming a small convolutional network trained end-to-end on raw pixels (92.6 %, macro-F1 0.921) while needing only 2.4 ms of feature extraction per image on a CPU. A feature-group ablation shows that region features from the segmentation stage are what separate off-centre and cluttered-background images. The dataset is derived from 300 real studio product photographs with five synthetically injected, fully labelled defect types. Code, data generator and all figures are released.**

***Index Terms*—image quality assessment, Laplacian variance, Otsu thresholding, morphology, connected components, random forest, e-commerce**

I. Introduction

Marketplaces such as Amazon, Flipkart and Myntra impose listing rules on product photographs: the item must be sharp, well exposed, centred, occupy most of the frame, and sit on a pure white background. Sellers upload millions of images a day, and manual review does not scale. An automatic inspector that *rejects early and explains why* saves review time and helps the seller re-shoot correctly. This is the problem behind reference project HSR-AIML-056 (*Automated E-commerce Image Quality and Compliance Inspector*).

Modern no-reference image-quality assessment (IQA) uses deep networks [3]–[5], but for this task the defects are low-level and well understood, so we ask a simpler question: *how far do classical DIP measures go, and does a learned model on top of them beat a learned model on raw pixels?* Our contributions are:

1) A reproducible dataset generator that takes real studio product photos and injects five labelled defect types (blur, noise, low contrast, cluttered background, off-centre framing).
2) A 10-dimensional DIP feature vector covering enhancement/contrast, filtering/noise, edge energy, thresholding, morphology and region analysis—each stage visualised.
3) A three-way comparison on a product-disjoint split: rule baseline vs. Random Forest on DIP features vs. a small CNN on pixels, with ablations and failure analysis.

II. Related Work

Variance of the Laplacian is a standard focus measure [1]; Immerkaer's and median-residual estimators are standard for additive noise [2]. Classical no-reference IQA such as BRISQUE [3] and NIQE [4] model natural-scene statistics; learned IQA such as KonIQ [5] trains CNNs on crowd-sourced opinion scores. Otsu's method [6] and morphological opening/closing [7] are the textbook tools for separating a product from a studio background. Our work differs in that it targets *marketplace compliance*—a multi-class defect decision rather than a scalar quality score—and keeps every stage interpretable.

III. Dataset

We start from the *Fashion Product Images (Small)* dataset [8], which contains studio photographs of apparel and accessories on a white background. We keep the first 300 images whose outer 1-pixel border has mean intensity above 235 (true white-background shots), pad them to a square and resize to 256×256. For every base image we generate six variants (Fig. 1):

- **clean**: unchanged;
- **blur**: Gaussian blur, kernel $k \in \{7, 9, 11, 13\}$;
- **noise**: additive Gaussian noise $\sigma \in [15, 35]$ plus 0.5–2 % salt-and-pepper;
- **low_contrast**: $I' = aI + b$, $a \in [0.25, 0.45]$, $b \in [100, 140]$;

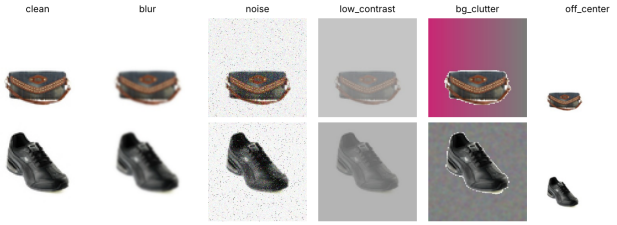


Fig. 1. Two base products and their six variants.

- **bg_clutter**: the white background (found by thresholding at 240 and closing) is replaced by a textured, gradient or striped background;
- **off_center**: the image is shrunk to 35–55 % and pasted into a random corner.

This yields $300 \times 6 = 1800$ JPEG images (quality 92) with exact labels. We split *by product*: all six variants of a product go to the same fold (70 % train = 1260, 30 % test = 540, 90 per class). Thus the test products are never seen during training, which prevents the classifiers from memorising product identity.

IV. Method

Fig. 2 shows the pipeline. Every image is converted to grey g; the ten features below are computed; a decision stage maps them to one of six classes. The binary PASS/FAIL verdict is *clean* vs. any defect.

A. DIP feature extraction

Sharpness (edge energy). $f_1 = \mathrm{Var}(\nabla^2 g)$, the variance of the 3×3 Laplacian response [1]. Blur removes high frequencies, so f_1 collapses (train-split median 243 for clean vs. 5.9 for blur).

Noise. $f_2 = 1.4826 \cdot \mathrm{MAD}(g - \mathrm{med}_3(g))$, the robust standard deviation of the residual after a 3×3 median filter. The median filter removes impulsive and Gaussian noise but keeps edges, so the residual is nearly pure noise. f_3 is the fraction of pixels at 0 or 255 that differ from their median by more than 100 (salt-and-pepper fraction).

Contrast (histogram). $f_4 = \sigma(g)$ and $f_5 = P_{99}(g) - P_1(g)$, the 1–99 percentile spread of the grey-level histogram. Fig. 4 shows that the low-contrast variant compresses the histogram to roughly a third of its width.

Background. f_6 is the fraction of pixels in an 8-pixel border frame brighter than 235 (border whiteness, the marketplace "white background" rule).

Segmentation and morphology. A foreground mask m is obtained by inverse Otsu thresholding [6], followed by a 5×5 morphological opening (removes speckle) and closing (fills holes), then keeping the largest connected component. $f_7 = \sigma(g \,|\, m = 0)$ measures background non-uniformity.

Region analysis. From the bounding box of m: f_8 = box area as a fraction of the frame, f_9 = distance of the box centre from the image centre (normalised), f_{10} = foreground fill ratio. These encode the framing rules ("product centred and filling the frame"). Fig. 3 shows every intermediate image for the six variants of one product; Fig. 5 shows how four features separate the classes.

TABLE I
Test set (540 images, 90 per class, product-disjoint). FRR = false reject rate on clean images; defect recall = fraction of defective images flagged.

Model	Acc.	Macro-F1	Defect rec.	FRR
Rule baseline (DIP)	0.913	0.915	0.991	0.111
Random Forest (DIP)	**0.983**	**0.983**	**0.993**	**0.033**
Small CNN (pixels)	0.926	0.921	0.960	0.033

TABLE II
Per-class F1 on the test set.

Model	clean	blur	noise	low_c	bg_cl	off_c
Rules	0.92	0.95	1.00	0.83	0.97	0.82
RF	0.97	0.99	1.00	0.96	0.99	0.99
CNN	0.89	0.97	0.77	0.97	0.93	0.99

B. Baseline: hand-tuned rules

Thresholds were chosen from the 5th/95th percentiles of the *training* split and applied in a fixed order: noise if $f_2 > 3$; low contrast if $f_5 < 100$; cluttered background if $f_6 < 0.3$; off-centre if $f_8 < 0.12$ and $f_9 > 0.25$; blur if $f_1 < 30$; otherwise clean. The order matters because noise inflates f_1 and a cluttered background lowers f_6 in the same way low contrast does.

C. Improved: Random Forest on DIP features

A Random Forest (300 trees, scikit-learn defaults) is trained on the same 10-D vector. It learns feature *interactions* and product-dependent thresholds that single global cut-offs cannot express, and gives feature importances for interpretation.

D. Learned alternative: small CNN on pixels

To check whether hand-designed features are worth it, we train a 4-block CNN (Conv-BN-ReLU-MaxPool, 16/32/64/64 channels, 96×96 RGB input, 2304-128-6 head, dropout 0.3) with Adam (10^{-3}), batch 64, 20 epochs, random horizontal flips. Training takes 65 s on two CPU cores.

V. Experiments and Results

Table I and Fig. 6 give the main comparison. The rule baseline already catches 99 % of defects but rejects 11 % of clean images, mostly because a single global sharpness threshold cannot cope with products that are intrinsically smooth (plain T-shirts, leather bags), and because dark, low-texture products shrink the Otsu mask and trigger the off-centre rule (Table II). The Random Forest on the *same* features removes most of these errors: macro-F1 rises from 0.915 to 0.983 and the false-reject rate drops to 3.3 %. The CNN on raw pixels, with no DIP prior, is worse than the RF (0.921) and in particular confuses noisy images with clean ones (noise F1 0.77); 96×96 down-sampling averages the noise away, which

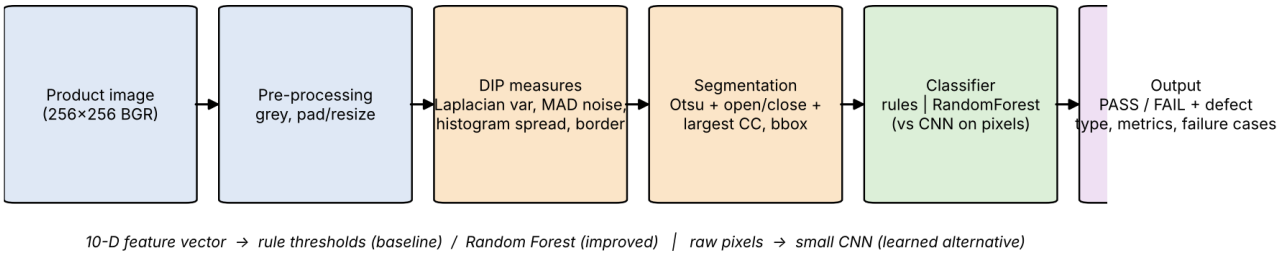


Fig. 2. Processing pipeline. Orange blocks are classical DIP stages; the green block is the decision stage that is compared in three forms.

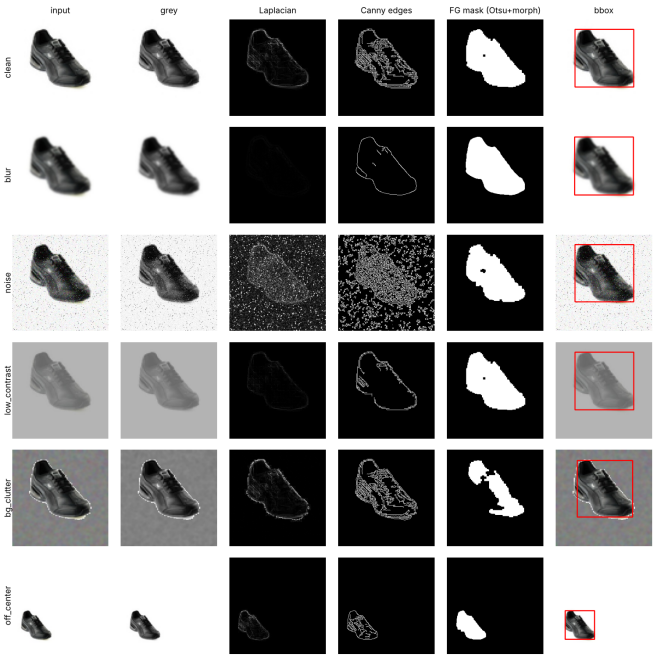


Fig. 3. Intermediate DIP outputs for the six variants of one test product: grey image, Laplacian magnitude, Canny edges, Otsu+morphology mask, bounding box.

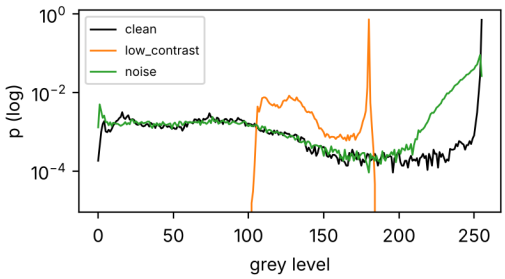


Fig. 4. Grey-level histograms (log scale) of the clean, low-contrast and noisy variants of the same product.

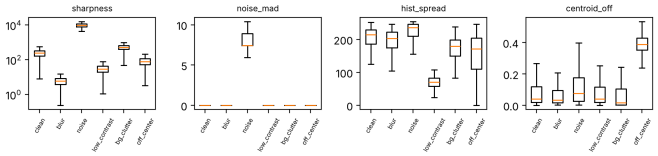


Fig. 5. Distribution of four features per class (whole dataset). Sharpness is on a log axis.

is exactly what the median-residual feature is designed not to do.

Feature importance. RF importances (Fig. 7) rank sharpness (0.31), border whiteness (0.19), centroid offset (0.10) and histogram spread (0.08) highest—one feature from each DIP stage.

Ablation. Table III adds feature groups cumulatively. Sharpness alone gives 0.73; adding noise and contrast features 0.94; adding background features 0.95; adding the region features from the segmentation stage 0.98. The segmentation stage is therefore necessary: without it, off-centre and cluttered-background images are not separable from clean ones. Removing the morphological opening/closing and largest-component step from the mask changed macro-F1 by $+0.004$, i.e. no measurable effect on this data: the studio backgrounds are nearly noise-free after JPEG compression, so Otsu alone already gives a clean mask. We keep the morphology for robustness to real uploads but report the result honestly.

Cost. Feature extraction takes 2.4 ms per 256×256 image on a CPU and RF training 0.5 s; the CNN needs 65 s to train and $\sim$3 ms per inference. The DIP+RF route is both more accurate and cheaper here.

A. *Failure analysis*

Fig. 8 shows representative errors. Three patterns recur. (i) *Smooth products*: a plain, uniformly coloured garment has little texture, so its Laplacian variance is as low as a blurred textured product; the rules call it blurred and even the RF errs on the smoothest ones. A product-adaptive sharpness measure (ratio of Laplacian energy before and after re-blurring) would address this. (ii) *Dark products on light-grey low-contrast images*: after the contrast compression the grey-level histogram still spans > 100 levels for very dark items, so the rule misses them; the RF recovers them through the border-whiteness fea-

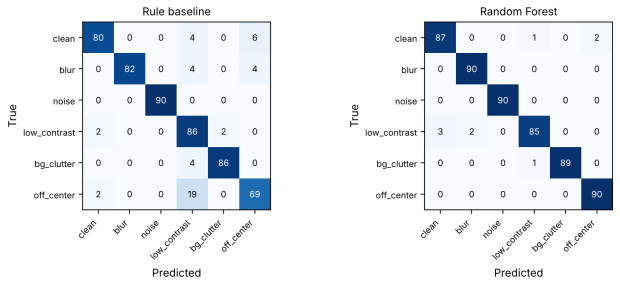


Fig. 6. Confusion matrices: rule baseline (left) and Random Forest (right).

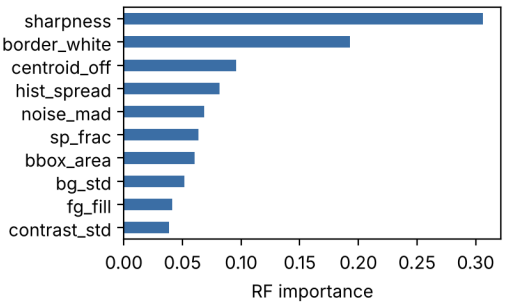


Fig. 7. Random-Forest feature importances.

ture ($f_6 \approx 0$ because the background became grey). (iii) *Small products that are legitimately centred* (jewellery, watches) have small bounding boxes and are occasionally called off-centre; the centroid offset condition reduces but does not eliminate this. The CNN's distinctive failure is noise vs. clean, discussed above.

VI. Conclusion

A ten-feature classical DIP pipeline—Laplacian sharpness, median-residual noise, histogram spread, border whiteness, Otsu+morphology segmentation and region statistics—combined with a Random Forest reaches 98.3 % accuracy on six-class e-commerce image compliance, beating both hand-tuned rules (91.3 %) and a pixel-level CNN (92.6 %) at a fraction of the cost. The ablation shows that each DIP stage contributes, with the segmentation/region stage essential for the framing rules. Limitations are that defects are synthetic and single-type per image; future work should test on real rejected uploads, multi-label defects and compare against BRISQUE/NIQE scores.

Reproducibility

Code (`src/make_dataset.py`, `dip_features.py`, `train_eval.py`, `ablation_and_diagram.py`), the generated dataset and all figures are in the submitted project folder. `python src/make_dataset.py && python src/train_eval.py` regenerates every number in this paper (seed 0).

TABLE III
Feature-group ablation (Random Forest, macro-F1).

Feature set	#feat	Macro-F1
sharpness only	1	0.731
+ noise, S&P, contrast, hist. spread	5	0.941
+ border whiteness, bg std	7	0.952
+ bbox area, centroid offset, fill (all)	10	**0.983**
all 10, mask without morphology/largest-CC	10	0.987

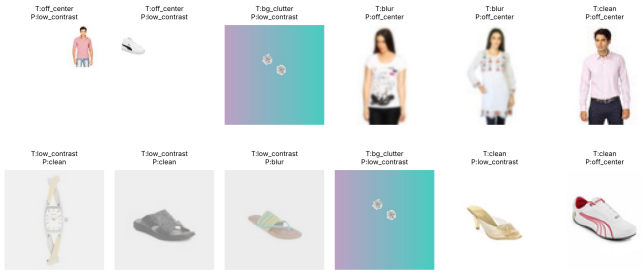


Fig. 8. Misclassified test images (T = true, P = predicted). Top: rule baseline. Bottom: Random Forest.

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